

English
Medium



EDU TERIA

72nd BPSC

Prelims Online Batch

BIOLOGY

LECTURE

9

Animal Tissue and Skeleton System



Animal Tissue and Skeleton System

Animal Tissue

In multicellular animals, similar cells group together to perform a specific function. This group of cells is called a tissue.

Structure may be similar or dissimilar. Based on structure and function, animal tissue can be classified as—

- I. **Epithelial Tissue**— Covering and protective tissue.
- II. **Connective Tissue**— Connect structure in the body, provides support to body system and provides medium of transformation of substances (food, or more gases etc.)
- III. **Muscle Tissue**— Helps in contraction and locomotion in the body.
- IV. **Nervous Tissue**— Generates and transport nerve impulses in whole body.

Animal Tissue

1. Epithelial Tissue:

- Covers body surfaces, lines cavities and organs.
- Cells tightly packed, little intercellular space.
- Dead layer of Cells.
- No blood supply
- Present in those membrane below which connective tissue is present.

Epithelial Tissue	
Simple epithelial tissue	Compound epithelial tissue
Single layer of Cell present	Two or more layers of Cell can be present.

Types:-

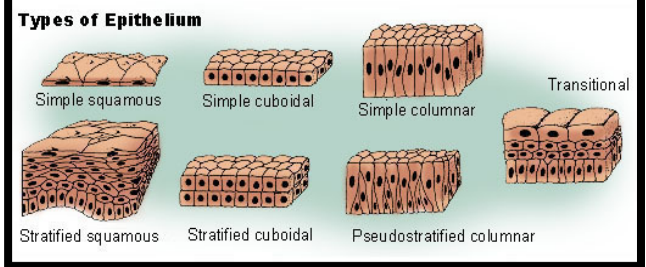
Squamous epithelium - thin, flat; helps in diffusion (lungs).

Cuboidal epithelium - cube-shaped; secretion & absorption.

Columnar epithelium - tall cells; absorption (intestine).

Ciliated epithelium - columnar with cilia (trachea, oviduct).

Glandular epithelium - secretion.



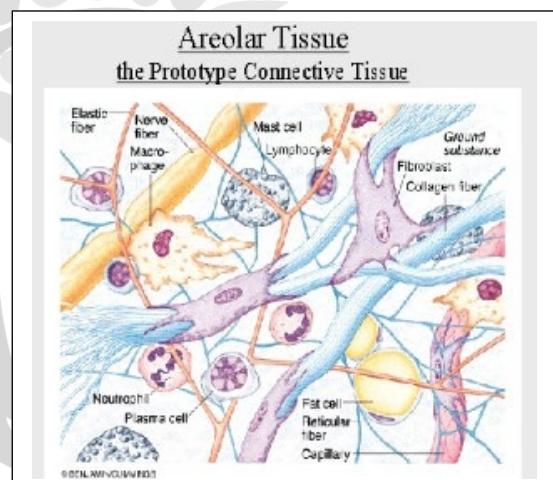
2. Connective Tissue:-

- Joins, supports, and transports materials.
- Cells loosely packed; large intercellular matrix.

Types:-

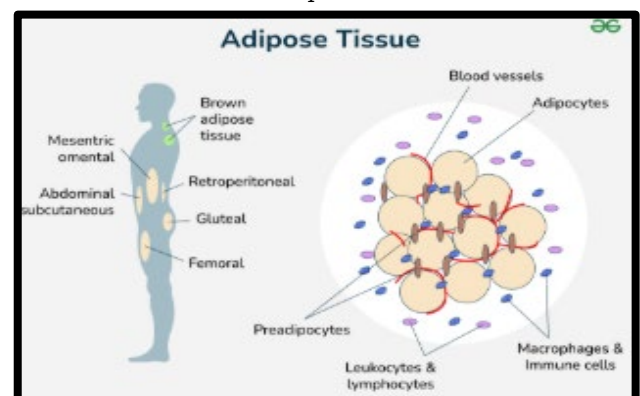
a) Areolar Tissue

- Between skin and muscles.
- Fills space, supports organs.



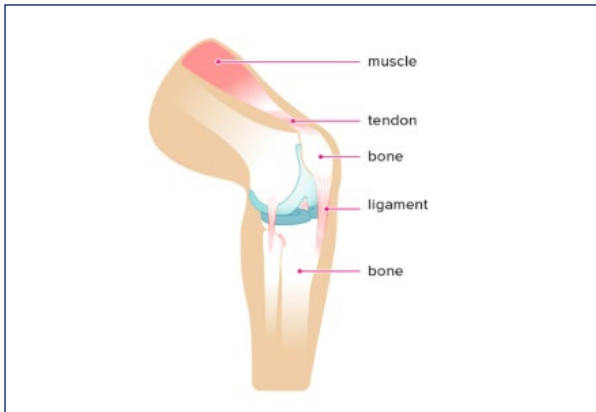
b) Adipose Tissue

- Stores fat, provides insulation.

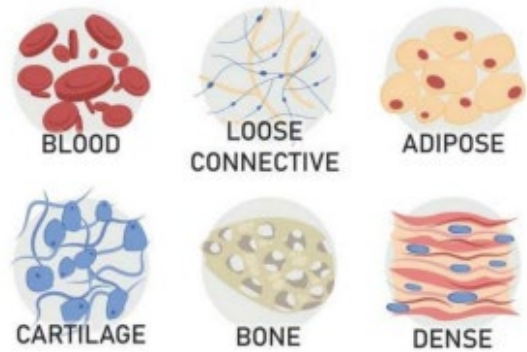


**c) Tendons & Ligaments**

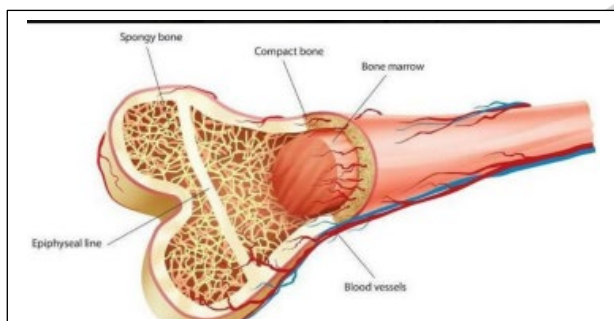
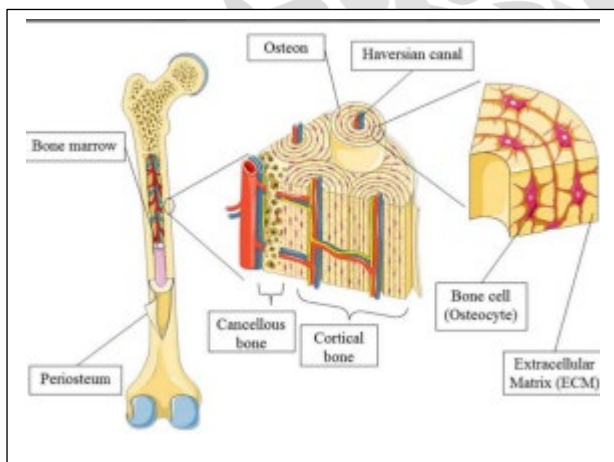
- **Tendons** - muscle to bone.
- **Ligaments** - bone to bone (more elastic).



- Transports gases, nutrients, waste.

TYPES OF CONNECTIVE TISSUE**d) Bone**

- Hard matrix with calcium and phosphorus.
- Supports body, protects organs.

**e) Cartilage**

- Flexible, less hard than bone.
- Present in nose, ear, joints,

f) Blood

- Fluid connective tissue.

3. Muscular Tissue

- Muscle tissue is responsible for movement of body parts. Responsible for movement.

Types:-**a) Striated (Skeletal) Muscles**

- Voluntary, striped, multinucleated.
- Moves bones.

b) Smooth Muscles

- Involuntary, spindle-shaped, no-striations.
- Present in alimentary canal, blood vessels.

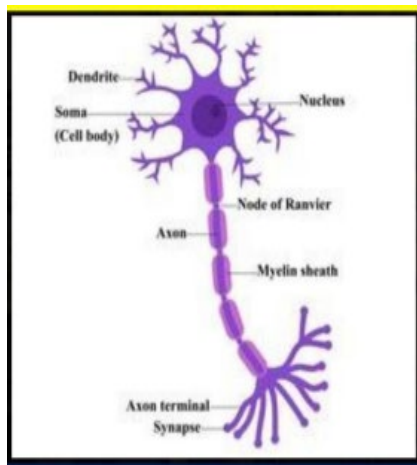
Skeletal muscle**Smooth muscle****Cardiac muscle****c) Cardiac Muscles**

- Involuntary, branched, striated.
- Found only in heart.

4. Nervous Tissue:-

- Made of neurons.
- Conducts impulses. SCIENCE Animal Tissue cell body
- Parts: Cell body, axon, dendrites.
- Controls and coordinates body activities.



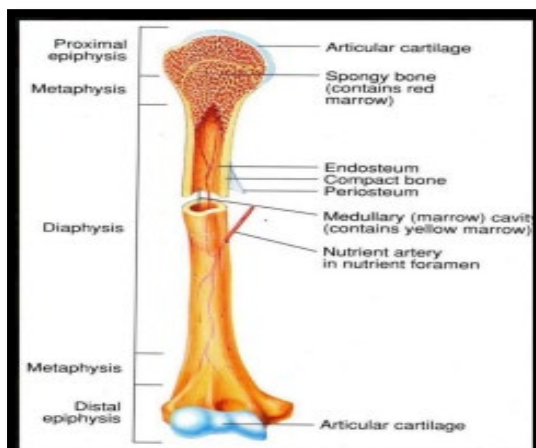


Human Skeletal system	
Axial skeleton	Appendicular skeleton
1. Skull	1. Fore limbs
2. Sternum	2. Hind limbs)
3. Vertebral Column	3. Pectoral girdle
4. Ribs	4. Pelvic girdle

Skeletal System

The skeletal system is the framework of bones and cartilage that supports and protects the body. In adult humans there are 206 bones.

Skeletal System	
Exoskeleton present outside of the body. - Dead in nature. - originate from ectoderm Eg- Nail horn shell (CaCO ₃) Hair Hoof	Endoskeleton present inside the body - living in nature. - Made up of mesoderm Eg.- Born Cartridge



Skull (Total 29 bone)

Cranial Bones	Facial Bone	Ears bones
• Frontal	• Mandible	Malleus(2)
• Parietal	• Vomer	Incus (2)
• Temporal	• Nasal	Stapes (2)
• Occipital	• Maxilla	
• Sphenoid	• Zygomatic	+ 1 neck bone (Hyoid bone)
• Ethmoid	• Lacrimal	
	• Palatine	
	• Inferior turbinals	

Human Skeletal System

Functions of the Skeletal System	
Function	Description
Support	Framework of body
Protection	Protect vital organs (Skull-brain, ribs—heart & lungs)
Movement	Provides attachment to muscles
Storage	Stores calcium, phosphorus and fats (in bone marrow)
Blood formation	Bone marrow produces RBCs, WBCs and platelets (Hematopoiesis)

Parts of Skeletal System

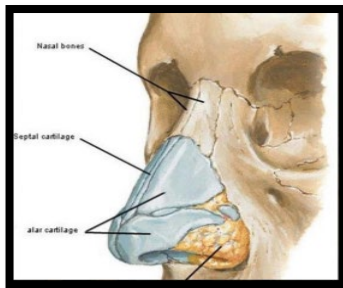
□ Human skeleton is divided into:

1. Axial Skeleton (80 bones)

- Skull → 29 bones
- Cranial bones - 8 bones
- Facial bones- 14
- Ear ossicles → 6 bones (malleus, incus, stapes)
- Hyoid bone - 1
- Vertebral column → 33 vertebrae (26 in adults after fusion)
- Rib cage → 24 ribs + 1 sternum

Vertebral Column Breakdown

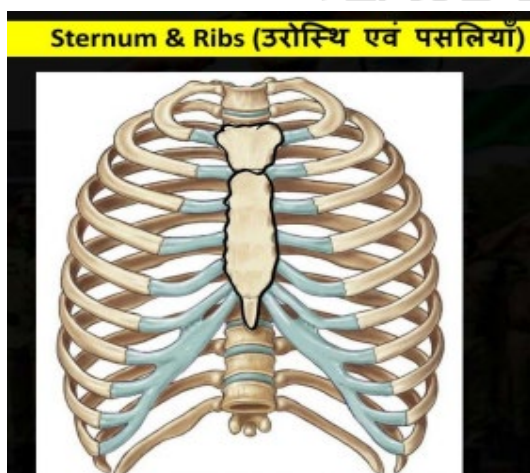
Region	Count
Cervical vertebrae	7
Thoracic Vertebrae	12
Lumbar Vertebrae	5
Sacrum	5 (fused)
Coccyx	4 (fused)



→ Cartilage

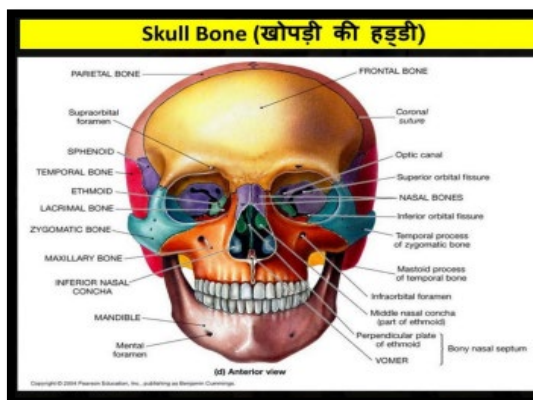
❑ Rib Cage

- 2 pairs of ribs
- 7 true ribs (1-7)
- 3 false ribs (8-10)
- 2 floating ribs (11-12)



2. Appendicular Skeleton (126 bones)

- Pectoral girdle (shoulders): 4 bones (2 clavicles + 2 scapula)
- Upper limbs: 60 bones
- Pelvic girdle (hips): (1+1) hip bones
- Lower limbs: 60 bones



Major Bones

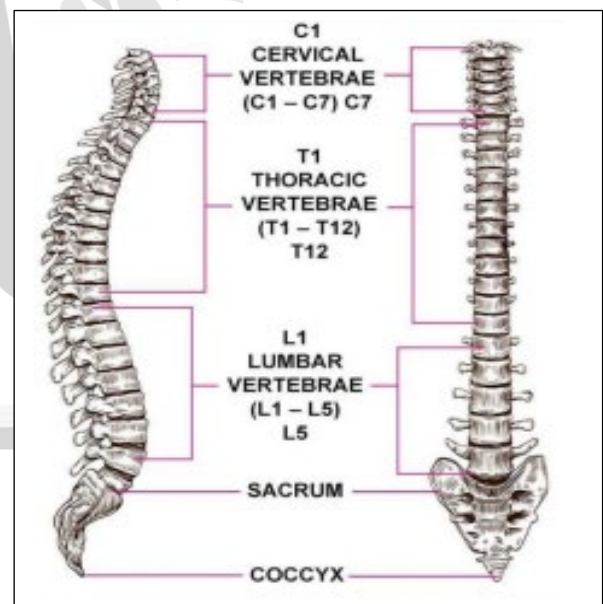
Bone	Fact
Femur	Longest and strongest bone
Stapes	Smallest bone (ear)
Skull bones	29 (8 cranial + 14 facial) + 6 ear bones + 1 neck bone.
Patella	Knee cap; largest sesamoid bone
Hyoid	Only bone not attached to any other bone

Bone Classification

Type of Bone	Example
Long bones	Femur, humerus
Short bones	Carpals, tarsals
Flat bones	Skull, sternum, rib
Irregular bones	Vertebrae
Sesamoid bones	Patella

Bone Tissue & Bone Marrow

- Compact bone: Hard outer layer
- Spongy bone: Contains spaces; found at bone ends
- Red marrow: Makes blood cells
- Yellow marrow: Stores fat



Vertebral Column



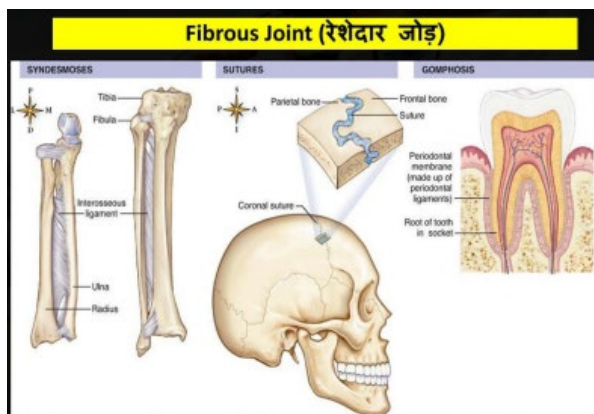
Joints

A joint is a point where two bones meet.

Types of Joints

- **Fixed joints** – skull
- **Hinge joints** – elbow, knee
- **Ball and socket joint** – shoulder, hip
- **Pivot joint** – neck
- **Gliding joint** – wrist

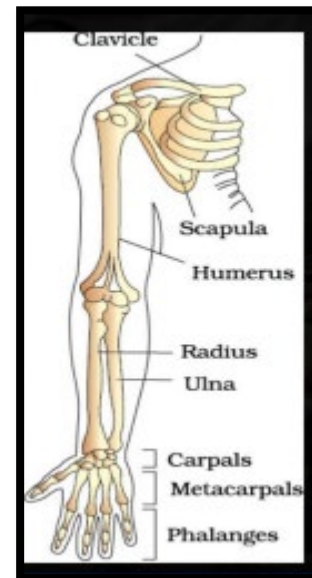
Joints		
Fibrous Joint	Cartilaginous joint	Synovial Joint



Cartilaginous joint:

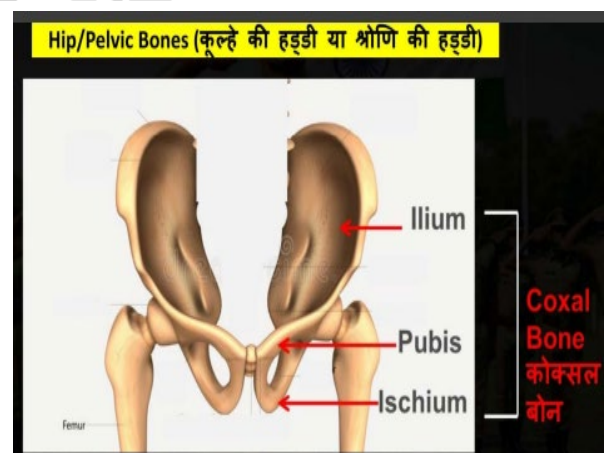
A **cartilaginous joint** is a type of joint where adjacent bones are united entirely by cartilage—either **hyaline cartilage** or **fibrocartilage**—rather than by fibrous tissue or a fluid-filled cavity.

Types of Synovial Joints	
Types	Examples
Ball-and-socket	Shoulder, hip
Hinge	Knee, elbow
Pivot	Neck (atlas-axis)
Gliding	Wrist bones
Saddle	Thumb joint
Condylod	Wrist joint



Fore limbs & Pectoral girdle

Disorders of Skeletal System	
Disorder	Description
Osteoporosis	Weak bones due to calcium loss
Arthritis	Joint inflammation
Rickets	Vitamin D deficiency in children
Fracture	Break in bone
Scoliosis	Abnormal spine curvature



□ Important Points

- Number of ribs: 12 pairs
- Jawbone: Mandible (only movable skull bone)
- Largest joint: Knee joint
- Bone protecting heart and lungs: Rib cage
- Bone protecting brain: Skull

